Answers on
Booklet - 4

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a)	g.p.e. to k.e. ✓	1	
)(i)	$v^2 = 2 \times 9.8 \times 2.8$ ✓ v = 7.4 (m s^{-1}) ✓ (g = 10 gives 7.48)	2	By energy change or by suvat
(ii)	neglecting/negligible/no air resistance ✓ (**all** gpe goes to ke) (constant 'a' **if** suvat approach)	1	**not** wind resistance

2.

(a)		**Use of s = ut + ½ at^2 i.e. s =1/2at^2** **t^2 = 2s / a related to y = mx**	C1 A1	with u = 0, allow the use of g in the equation e.g. gradient = 2 / a
(b)	(i)	**(gradient = using correct values from graph)** **= 0.20**	C1 A1	 Allow 0.19 to 0.21 allow 0.2 (1 sf) inverse gradient scores zero
	(ii)	**s = ut + ½ at^2 with substitution of points read from graph.** **Or t^2 = 2s/a, therefore gradient = 2/a** **a = 2 / gradient** **= 2 / 0.20 = 10.0 (m s^{-2}) allow 10**	 M1 A1	 note ecf from(b)(i) but not (a)
(c)		**width of the ball / end error in readings from the ruler / time delay due to electromagnet or release of ball or trapdoor / air resistance**	Max B2	no credit for any random errors in readings

3.

(a)(i)	a = 1.8 / 0.8	C1
	= 2.25 (m s^{-2})	A1
(ii)	F = ma	
	= 0.8 x 2.25	C1
	= 1.8 (N)	A1
(iii)	area under the graph or s = (u + v) t / 2 etc	C1
	distance = (1.8 x 0.8) / 2	M1
	hence distance = 0.72 m	A0
(b)(i)	distance BC = 2 – 0.72 = 1.28 (m)	C1
	time = 1.28 / 1.8	C1
	time = 0.71 (s)	A1
(ii)	straight horizontal line of at least 2 small squares	B1
	until 1.5 (s) [note ecf from (b)(i) i.e. 0.8 + (b)(i)] within ½ square	B1
	steep line (do not consider shape) to zero speed (max of two squares)	B1

4.

(a)	W = 75 x 9.8 ✓ = 735 N ✓	
	(accept (75 x 10) = 750 or (75 x 9.81) = 735.75)	2
(b)(i)	mention air resistance/drag force ✓	
	air resistance/drag … increases with speed ✓	
	resultant/accelerating force decreases ✓	3
(ii)	the idea of balanced forces/resultant force zero ✓	
	(allow 'forces equal')	1
(c)(i)	air resistance/drag increases ✓.....	
	.. idea of increased area / unbalanced force upwards✓	2
(ii)	for the idea that	
	'the speed at which air resistance = weight is less/different' ✓✓	
	(<u>or</u> 'he has slowed down to a speed when drag = weight'	
	<u>or</u> ' once parachute opens the forces again 'level out' reaching	
	equilibrium but at a different speed')	2
	(for just 'forces become balanced again' 1 mark)	

5.

(a)		Arrows of the correct length ± 2 mm	B1
(b)	(i)	At B: zero / less than / smaller	B1
	(ii)	At C: smaller <u>and</u> opposite / negative direction	B1
	(iii)	At D: same size / no change and no change for all three horizontal components	B1
(c)		No change in horizontal component since no air resistance / no components of forces in this direction	B1
		At B maximum height (therefore no vertical component)	B1
		At C direction of motion has changed / falling back down / vertical component increases as acceleration is vertically down	B1
		At D(velocity is same magnitude but opposite direction to starting velocity) ball has accelerated (due to gravity) down to original speed / idea of time or distance down being the same as that going up	B1

6

(a)(i)	horizontal velocity = 25cos30	B1
	= 21.65 (m s^{-1})	A0
(ii)	vertical velocity = 25sin30	C1
	= 12.5 (m s^{-1}) allow 13 (2 sig. fig.)	A1
(b)(i)	straight line with positive gradient through 0,0	
b(ii)	curved line with positive increasing gradient zero gradient at 0.0	
	MAX 3	B3
(c)(i)	straight line with positive gradient (through 0,0)	B1
	line to time axis when ball hits sand	B1
	non vertical line coming down to zero between impact and stone at rest	B1
(ii)	horizontal line non zero (positive value)	B1
	negative (opposite) value from impact with sand	M1
	returning to zero between impact and stone at rest	A1
		MAX 5

7.

(a)(i)	speed = 1800 / 65	C1
	= 27.7 (m s^{-1}) / 28 (m s^{-1})	A1
(ii)	loss in potential energy = m x g x h	C1
	= 495000 (J)	A1
(b)(i)	work done = 250 x 1800	C1
	= 450000 (J)	A1
(ii)	1. Kinetic energy gained = 495000 – 450000	
	= 45000 (J)	B1
	2. k.e. = ½ mv^2 (allow one mark if seen in 1 or 2) Possible ECF.	C1
	v^2 = 45000 x 9.8 x 2 / 900 (allow g = 10NKg^{-1} here)	C1
	v = 31.3 (m s^{-1})	A1
(iii)	greater final speed	M1
	as the skier will also do some work (pushing with poles)	A1

8.

(i)	g.p.e. = mgh	C1
	60 x 80 x 9.81 x 900	
	42379200 (allow use of 9.8 and answer of 42336000) (J)	
	(42 MJ)	A1
(ii)	total energy input = 6800 x 9.81 x 900	
	or 42379200 + 2000 x 9.81 x 900	C1
	power = 60037200 / (5 x 60)	
	or (42379200 + 17658000) / (5 x 60)	C1
	= 200124 W (200kW) (199920 using 9.8)	A1
	unit penalty	-1

9.

(a)(i)	thinking distance = 25 x 0.65 = 16(.25)	C1
	stopping distance = 40 + 16.25 = 56(.25) (m)	A1
(ii)	Ke = ½ x 800 $(25)^2$	C1
	= 250 x 10^3 (J)	A1
(iii)	work done (by braking force) = loss in Ke / $(25)^2$ = 0 + 2a x 40	C1
	F x 40 = 250000 / a = 7.81	C1
	F = 6250 (N)	A1

10.

(a)	(i)	Distance travelled from the instant the driver sees the problem until the brakes are applied	B1
	(ii)	Distance travelled by car after the brakes are applied and the car has <u>come to rest</u>	B1
(b)	(i)	Distance = 25 x 0.62	C1
		= 15.5 (m)	A1
	(ii)	F =ma	B1
		Force down slope = Wsin5.8 / 843	B1
		B – Wsin5.8 = ma	B1
		a = (5520 – [850 x 9.81 x sin 5.8]) / 850	B1
		= 5.50 m s^{-2}	Max B3 A0
	(iii)	s = u^2 / 2a / = $(25)^2$ / 2 x 5.5	C1
		= 56.8 m	A1

11.

i	Dramatic change(s) in <u>density</u> (at 3.0 Mm and 5.1 Mm) (AW)	B1	**Not**: There are three (distinct) layers / Each layer has different density
ii	mass = $0.18 \times 6.0 \times 10^{24}$ (=1.08×10^{24} kg) or radius = 1.3×10^{6} (m) volume = $\frac{4}{3}\pi \times (1.3\times10^{6})^{3}$ density = $\frac{1.08\times10^{24}}{9.20\times10^{18}}$ density = 1.2×10^{5} (kg m^{-3})	C1 C1 A1	**Note**: The first C1 mark is for determining the mass or the radius of core Possible 10^{n} errors **Bald** answer of 1.2×10^{5} (kg m^{-3}) or 1.17×10^{5} (kg m^{-3}) scores 3 marks **Allow**: 2 marks for $\frac{6.0\times10^{24}}{9.20\times10^{18}} = 6.5\times10^{5}$ (factor of 0.18 missed out) **Note**: The <u>last</u> <u>two</u> C1 and A1 marks cannot be scored if incorrect radius is used. Hence no further marks for $\frac{1.08\times10^{24}}{\frac{4}{3}\pi\times(6.4\times10^{6})^{3}}$ or $\frac{1.08\times10^{24}}{\frac{4}{3}\pi\times(5.1\times10^{6})^{3}}$, etc

12.

a		Any <u>two</u> from: • area • speed / velocity • viscosity (of air) / temperature / density • (surface) texture / 'aerodynamic' (shape)	B1×2	**Not**: shape / size **Allow**: 'streamlining'
b	i	Correct <u>directions</u> of arrows *W* and *D*	B1	Award the mark for two arrows in opposite directions as long as <u>one</u> of them is labelled
	ii	weight = 75×9.81 weight = 736 (N) or 740 (N)	 B1	**Reminder**: weight can be quoted to more than 2 sf (e.g: 735.75) **Not**: '$75 \times 10 = 750$ N'
	iii	$D = 0.30 \times 20^{2}$ (= 120 N) $736 - 120 = 75a$ $a = 8.2$ (m s^{-2})	C1 C1 A1	 **Allow**: Answer to 2sf or more **Bald** answer of 8.2 or 8.21 scores 3 marks **Note**: Using 740 (N) gives an answer 8.3 (m s^{-2})
	iv	(*D* and *W* are) equal	B1	**Not**: *D* and *W* are 'balanced/equilibrium'
	v	drag = weight $736 = 0.30 \times v^{2}$ $v = 49.5$ (m s^{-1}) or 50 (m s^{-1})	 C1 A1	 **Bald** answer of 49.5 (m s^{-1}) or 50 (m s^{-1}) scores 2 marks

Measurements:		
height (of wall)	B1	**Allow:** 'distance (of fall)' instead of 'height'
time (of fall)	B1	
Instruments:		
ruler / tape (measure)	B1	
stopwatch / timer / clock /video	B1	✎ **The 4th B1 can only be scored if** ***stopwatch / timer / clock / video*** **(camera) is spelled correctly**
$g = \frac{2s}{t^2}$ / $g = 2 \times$ gradient of s-t^2 graph	B1	**Allow:** Use of 'a' instead of 'g' **Note**: a must be the subject
Note: Allow full credit if candidate has used alternative approaches using $v^2 = u^2 + 2as$ or $v = u + at$.		
Any <u>two</u> from: g is an estimate because • air resistance / drag ignored • parallax problems with 'landing time' • starting / stopping the clock	B1×2	**Allow:** 'wind resistance'/'resistive force' for first bullet point **Allow**: 'reaction time' but **not** 'human error' for the third bullet point

13.